

Benchmark Report: yolov8s_seg_metal_nut

1. Model

- ONNX: `model/yolov8s_seg_metal_nut.onnx`
- Architecture: YOLOv8-seg (single class, det-only at runtime)
- Input: `images` [1,3,640,640], BGR->RGB, /255
- Outputs (engine binds both, parser reads only `output0`):
- `output0` [1,37,8400] = 4 bbox + 1 cls + 32 mask coef
- `output1` [1,32,160,160] = prototype masks (UNUSED)

2. Hardware

- Target: GCP VM `aoi-d5` (us-central1-a), L4 GPU
- DeepStream 9.0 container

3. Engine Build

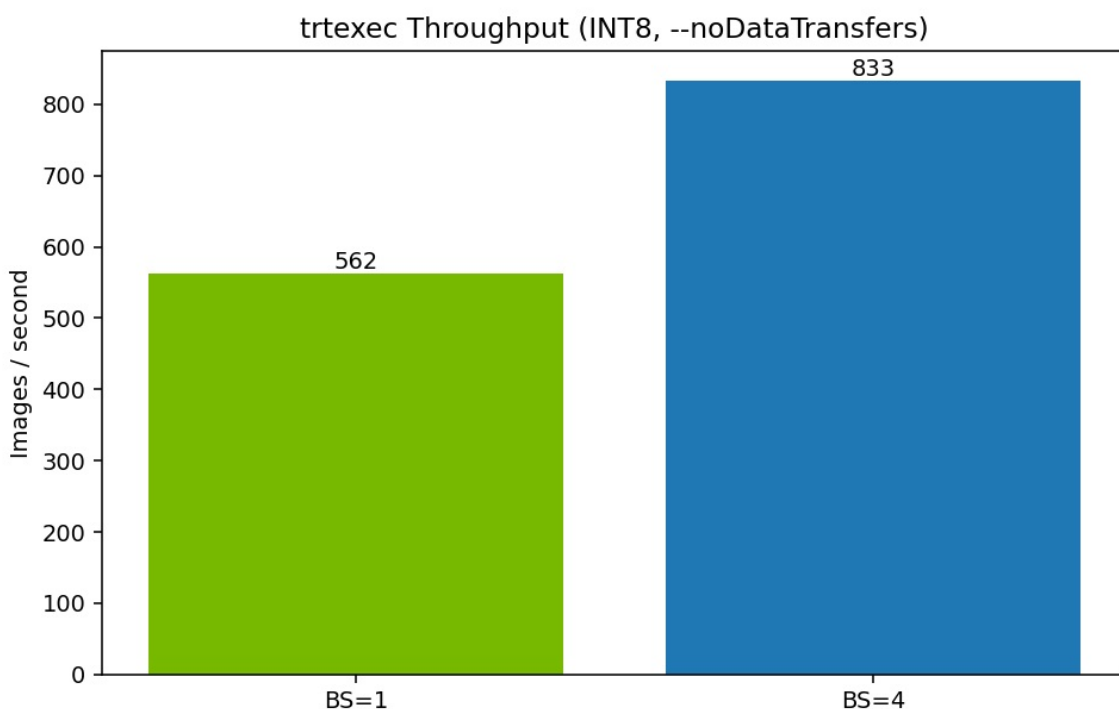
- Precision: INT8 (`metal_nut_int8.cache`)
- Dynamic batch [1..4]
- Engine: `benchmarks/engines/yolov8s_seg_metal_nut_dynamic_b4.engine`

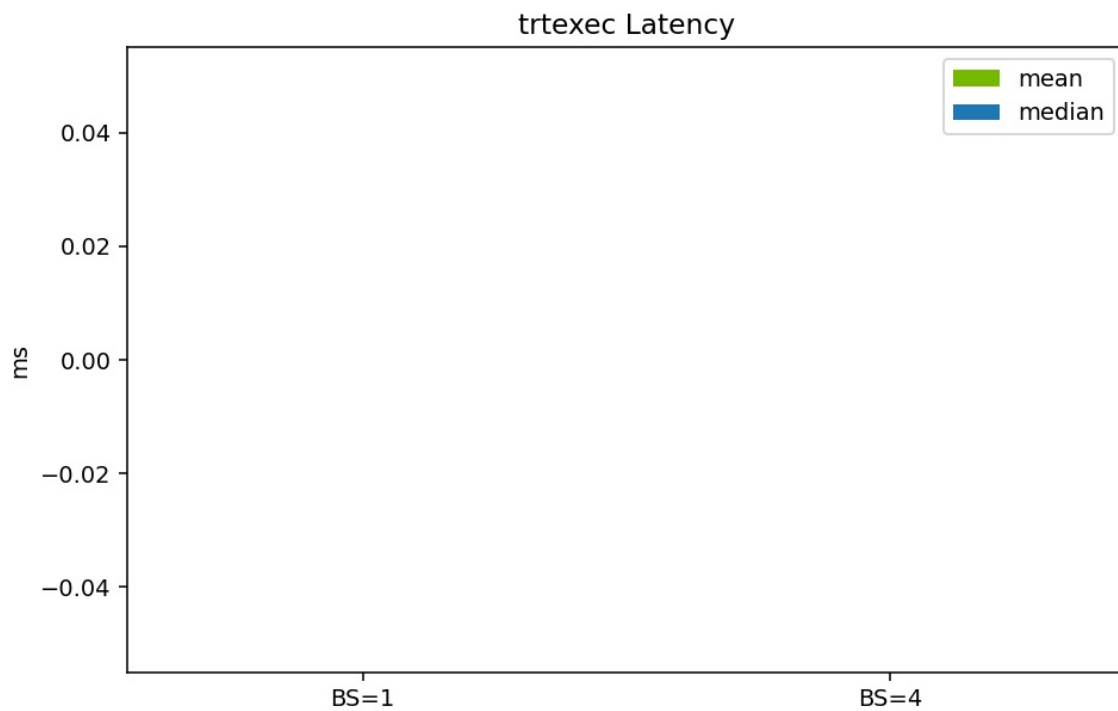
4. trtexec BS=1

- Throughput: 562.12 qps
- Latency mean / median: 0.00 / 0.00 ms

5. trtexec BS=4

- Throughput: 208.17 batches/s = 832.68 img/s
- Latency mean / median: 0.00 / 0.00 ms

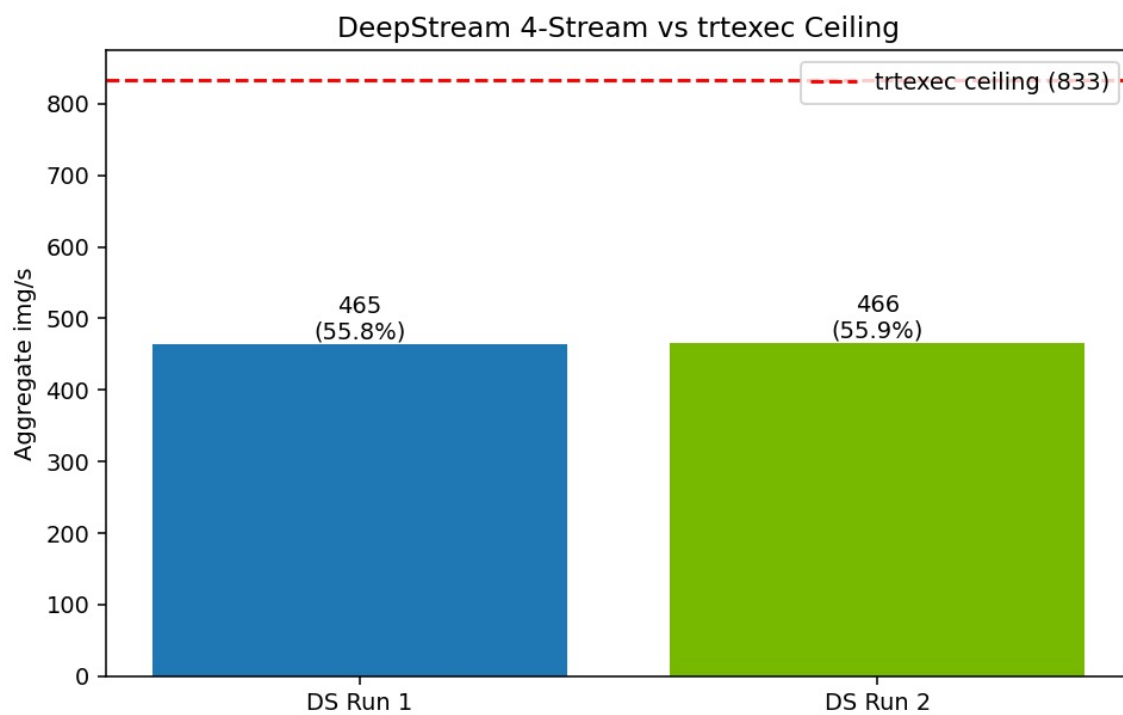


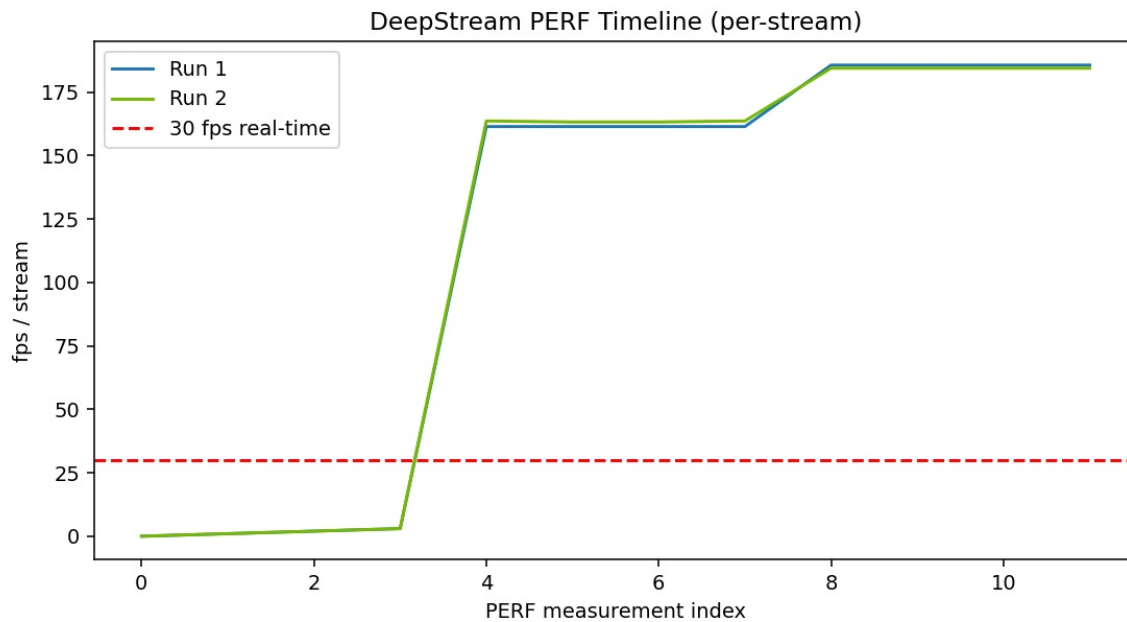


6. DeepStream Multi-Stream (batch=4, 4 streams)

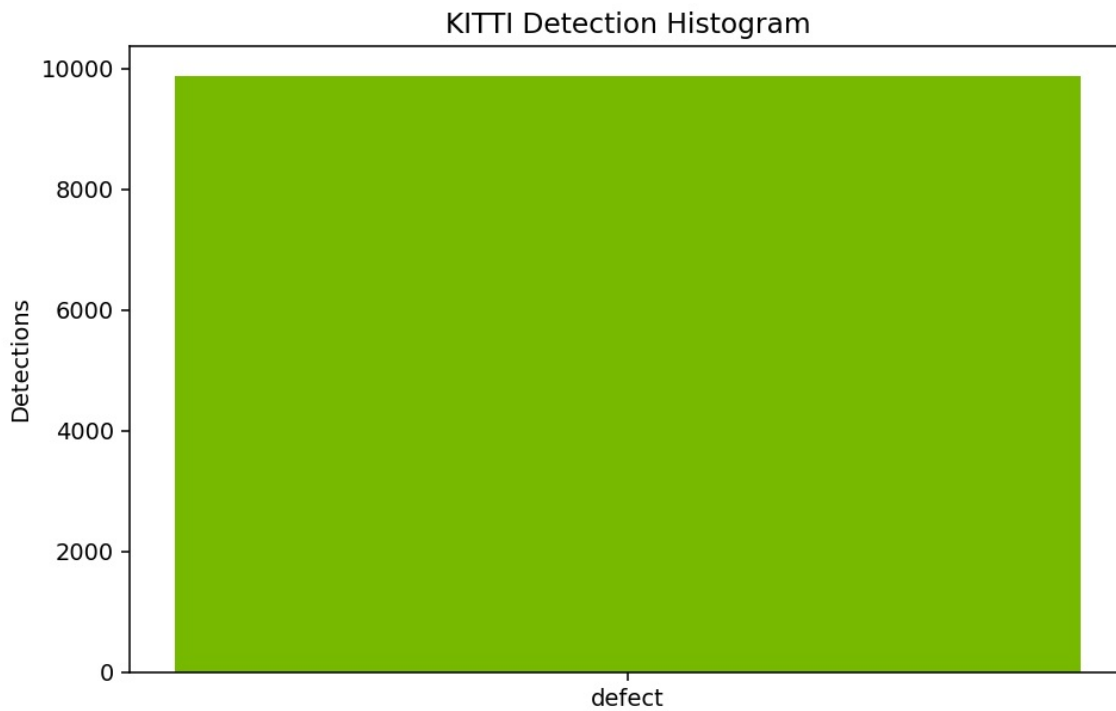
Run	Streams	fps/stream	total img/s	DS efficiency
Run 1	4	116.17	464.68	55.8%
Run 2	4	116.43	465.72	55.9%

Real-time @ 30 fps/stream: **YES**





7. Detection Sanity (KITTI dump)



8. Pipeline Stages

1. Engine build (`scripts/build_engine.sh`)
2. Loop sources (`scripts/prep_test_loops.sh`)
3. Single-stream mock factory (`scripts/run_pyservicemaker_mock_factory.py`)
4. Multi-stream bench (`scripts/bench_multistream.sh`)
5. Report (this script)

9. Parser Notes

- Det-only: 32 mask coefficients in `output0` channels 5..36 are skipped.
- `output1` (prototype masks) bound but not consumed.
- `cluster-mode=2` (DS NMS), `pre-cluster-threshold=0.25` .

10. Known Limitations

- No mask metadata produced; instance segmentation requires `network-type=3` plus a sigmoid+matmul postprocess that this skill does not cover.
- INT8 calibration cache reused as-is; rebuild if class distribution shifts.

11. Encoder Mode

- Primary: `nvv4l2h264enc -> .mp4`
- Fallback: `theoraenc + oggmux -> .ogv`

12. Files

- Engine: `benchmarks/engines/yolov8s_seg_metal_nut_dynamic_b4.engine`
- Logs: `benchmarks/b1/trtexec_b1.log`, `benchmarks/b4/trtexec_b4.log`, `benchmarks/ds/ds_s4_run1.log`, `benchmarks/ds/ds_s4_run2.log`
- Sample video: `samples/metal_nut_mock.{mp4,ogv}`